Case 2 - Medical AI Adverse-Event Investigation

CONFIDENTIAL - SYNTHETIC BOARD / TECHNICAL REVIEW MATERIAL

Synthetic confidential incident package designed to force causal reasoning across model behavior, software, clinical policy, deployment governance, and contractual responsibility.

Board incident-review context

Aster Vale Health develops an adaptive sepsis deterioration model deployed at Northbridge Medical Center. A serious adverse event occurred after release 4.8. The model executed the configured policy as designed; the open question is whether the failure originated in the model, software implementation, clinical policy, deployment controls, or human use.

1. Cleared / approved operating envelope

Item	Documented requirement
Intended use	Risk stratification support for adult inpatients; not a stand-alone diagnostic.
Alert policy	High-risk alert when calibrated 6-hour deterioration probability >=0.35.
Adaptive update	Monthly recalibration of intercept and calibration slope permitted; feature set and clinical action policy locked.
Override	Hospital may suppress alerts only through validated unit-level configuration with clinical governance approval.

2. Hospital clinical policy excerpt

Northbridge policy NMC-SEP-17, revision 2024-11, instructs nursing staff to defer escalation when an alert occurs within four hours after initiation of a documented fluid-resuscitation protocol unless specified instability criteria are present. In March 2026, the hospital's sepsis committee revised its fluid protocol and shortened reassessment to two hours, but NMC-SEP-17 was not updated.

3. Release 4.8 software notes

Change	Status
Monthly calibration refresh	Implemented; no feature or threshold changes.
Alert suppression rule	Reads hospital-configured suppression interval from policy service.
Fallback if policy service unavailable	Uses last validated configuration.
New self-learning feature weights	Not implemented.

4. Incident timeline

Time	Event
08:12	Patient begins revised fluid protocol.
10:31	Model score rises to 0.47; high-risk condition satisfied.
10:31	Policy service returns suppression_interval=4h from NMC-SEP-17.
10:31	Software records ALERT_SUPPRESSED_POLICY; no clinician-facing alert.
11:04	Lactate and blood pressure worsen.
11:58	Bedside nurse escalates based on clinical signs.
12:17	Rapid-response team arrives.

5. Model and QA evidence

Offline replay using release 4.7, release 4.8, and a frozen pre-adaptation model all cross the 0.35 alert threshold between 10:27 and 10:33. Independent code review finds the suppression function implemented exactly as specified. The March clinical-policy revision was approved in committee minutes but never entered into the policy-service change-control queue.

6. Internal correspondence

Feb 18 - Product manager: 'Hospital policy service is authoritative for local workflow rules; we should not hard-code these.' Mar 22 - Clinical integration lead: 'Northbridge says the new fluid protocol is live. I don't see a corresponding SEP-17 ticket; assuming their governance team owns that.' Apr 03 - Hospital analyst: 'We updated the bedside protocol but I wasn't aware the AI suppression interval was sourced from the older policy object.'

7. Contract / responsibility excerpt

Vendor is responsible for validated operation of the software and documented model-update mechanism. Hospital is responsible for clinical workflow policies, local configuration approval, and timely notification of policy changes affecting integration behavior. Vendor must provide reasonable safeguards against configuration inconsistency where technically feasible.

Challenge prompt

> **Reconstruct the adverse event from this confidential package. Separate model failure, software failure, clinical-policy failure, deployment/change-control failure, and human error. Identify the evidence for each conclusion, unresolved facts that could change the analysis, and the highest-priority corrective controls. Do not infer legal liability merely from contractual responsibility.**